

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457849.709 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.018
Transit depth (Rp/Rs)^2	2.82 +/- 0.62 [%]
Orbital Inclination (inc)	85.42 +/- 1.37
Airmass coefficient 1 (a1)	1.164 +/- 0.024
Airmass coefficient 2 (a2)	-0.104 +/- 0.033
Scatter in the residuals of the lightcurve fit is	2.03 %
Best Comparison Star	None
Optimal Aperture	3.71
Optimal Annulus	11.1
Transit Duration (day)	0.0588 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.168 $\pm$ 0.018 vs 0.1594 (deviation 0.37 σ)
Duration fit vs published	0.0588 d vs 0.0512 d (deviation 2.33 σ)
Mid-time O-C	-1.8 min
Depth SNR	7.2
Residual scatter	2.03 %

Light curve

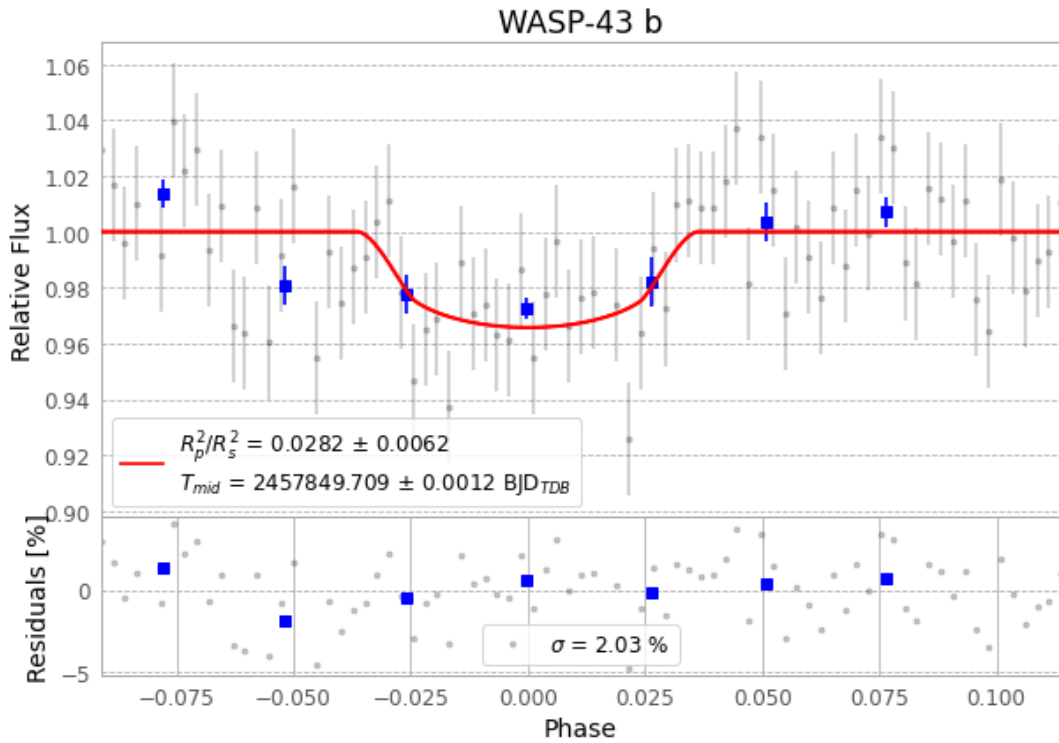


Figure 1 — FinalLightCurve_WASP-43 b_2017-04-05.png (SHA-256 359c03324cd9194f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [391, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ecaa0f8bb5451797...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

81 FITS frames (53 MB), WASP-43170406030612.FITS → WASP-43170406070612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170406.log (0652991b2de9...), FinalLightCurve_WASP-43 b_2017-04-05.png (359c03324cd9...), FinalLightCurve_WASP-43 b_2017-04-05.csv (c6a443d9b902...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 18:47Z.